The Digital Shift: Analyzing Trends in Book Publishing and Library Visits

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***Abstract*—Our study examines the changing role of public libraries among the increasing number of online publishing platforms, with a narrowed scope focusing on the circulation trends within Texas. Building from the study, "Kindle's Shadow: How Digitization Dims Novelty in Creative Industries" by Khwan Kim, which explains that digital platforms often popularize the same mainstream content at the expense of creative innovation, our research investigates whether similar dynamics are mirrored in patterns found from the rise of digital reading content and in-person library visitation per year. However, while Kim explores how the recommendation systems' algorithms restrict creative innovation, our research pivots the focus onto the consumer side and analyzes how the growing digital platforms are influencing public library usage.**

Leveraging circulation data from the Texas State Library and Archives Commission, we conducted a comparative analysis of physical versus digital checkouts, alongside in-person library visits per year. Our findings have revealed an incline in digital borrowing, a slight decline in physical foot traffic, and a growing trend towards hybrid models. These results suggest that many public libraries are already aware of the growing demand for digital content and are actively adapting to the online information environment. This study conducts research though both the Texas library and digital publishing statistics to better understand how public reading habits are changing in the increasingly digital age. It shows that as more people turn to digital formats like eBooks and audiobooks, libraries must update their services and technology to meet these new needs.

***Index Terms*—Public Libraries, Library Information, Library Analytics, Digital Publishing, Digital Circulation, eBooks, Kindle, Audiobook, Library Usage Trends, Machine Learning, Random Forest, XGBoost, Time Series Analysis.**

I. Introduction

Public libraries have always served as places offering access to physical and digital resources, educational programs, and quiet spaces to read or work. This has questioned traditional libraries, especially in this increasingly digital era of quick information access.

Over this past decade, digital publishing has grown significantly. Ebooks allow readers access to huge catalogs of books at the touch of a few buttons. This raises a few concerns about whether this digital shift will work together with libraries or if it will compete with public library usage.

Additionally, public libraries in some regions must also navigate declining foot traffic, inconsistent funding, and differences among readers in how they use library resources. These challenges showcase the need to understand what libraries are doing to adapt to the growing number of digital competitors and whether any these changes align with public usage patterns in order to determine what improvements would help libraries stay relevant to the general public.

As such, our research aims to identify the relationship between digital publishing trends, from Kindle sales data, physical publishing and library visitations trends found in Texas public library datasets from 2018 to 2023. By combining library statistics with consumer-focused Kindle sales data, this study offers a better view of how digital content usage affects reader-based physical and digital library services. Through detailed data analytics and machine learning techniques, this research also attempts to forecast future trends and uncover

patterns to help make informed decisions about future library resource allocation, digital transition strategies, and access to reading materials across different formats.

II. Literature Review

The foundation of this project is "Kindle's Shadow: How Digitization Dims Novelty in Creative Industries" by Khwan Kim. The paper by Kim, 2024 tests how the implementation of digital platforms, especially the Kindle ecosystem in Amazon, affects the novelty of new books compared to previously published works and the extent to which popular authors' new works are different compared to their previous works. Kim found that although digitization has expanded the number of books and the number of authors, it has unfortunately also led to a decline in creativity. This explains how when one type of work becomes popular, other authors try to mimic what works instead of branching out and creating something new, in part because that way their new works have a better chance at being pushed to readers. This is not solely the fault of authors, as Kim clarifies that this is a design feature of the digital platforms themselves. User reviews, star ratings, and recommendations algorithms all work together and seem to promote mainstream and similar content over obscure, experimental or otherwise niche works before those new stories become popular Kim, 2024.

Inspired by Kim's findings, our research asks: What do trends in yearly digital and physical circulation of books indicate about public interest? Given the difficulty of quantitatively measuring literary novelty in circulation data, we pivoted to analyzing user behavior, more specifically, focusing on how the growth of digital formats (eBooks, Kindle content, and audiobooks) has impacted physical library visits. Therefore, to begin to learn more about the context surrounding this topic, we drew upon a wide range of literature.

As such, we found that a study by Krizhevsky, Sutskever, and Hinton pioneered a convolutional neural network model that revolutionized image recognition. While their work was not focused on libraries, it established the deep learning techniques that now power digital recommendation engines and categorization tools within library websites Krizhevsky et al., 2012. Similarly, another paper by Zhang, Wang, and Liu further advanced this field through a detailed review of deep learning within sentiment analysis, providing insights into how AI interprets user reviews, emotions, and preferences, elements increasingly used to guide content curation in digital environments Zhang et al., 2018.

However, in recent years, a pandemic swept across the world. This had a huge impact on physical activities, including visiting a library. Therefore, we looked into a study by Ćirić and Ćirić that observed the surge in digital engagement during the COVID-19 pandemic, which highlighted the importance of resilience in library services. Their findings emphasize that external shocks can accelerate digital adoption, though they do not necessarily replace physical formats Ćirić and Ćirić, 2021.

But, turning the focus back to more library-specific research, a paper by Duncan conducted a deep dive into the growing demand for eBooks and the importance of libraries adapting to technological shifts. He identified both infrastructural challenges and the need for community education around new formats Duncan, 2010. In the same way, a study by Yuan, Van Ballegooie, and Robertson investigated user preferences between print and digital formats, noting that while digital books offer convenience, many readers still prefer physical books for deeper engagement, a duality that continues to shape collection development strategies Yuan et al., 2018. Likewise, another paper by Blummer and Kenton who systematically reviewed eBook usage in academic libraries, found steady increases in digital access alongside ongoing challenges in licensing, discoverability, and digital literacy Blummer and Kenton, 2020. Issues, that are relevant to public libraries as well. Similarly, a study by Matthews and Johnson also analyzes academic libraries, and they found that, for various factors, first generation college students tend to prefer the printed version of a book over the book's digital counterpart. They found that this method of studying helps those students in retaining the book's content for classes Matthews and Johnson, 2023.

Furthermore, a great study by Palmer Palmer, 2022 offered a forward-looking perspective on how libraries are evolving into hybrid community learning spaces. Her analysis reinforces the need for strategic digital integration to ensure long-term relevance and offers a potential look into what has been working in some libraries.

Lastly, a paper by Spjeldnæs and Karlsen Spjeldnæs and Karlsen, 2024 examined how digital reading affects leisure habits, suggesting that while convenience increases, the emotional and cognitive experience of reading may change, potentially influencing user satisfaction and long-term behavior.

Together, all these works offer a robust framework for examining how digitization, user behavior, and institutional priorities intersect. By conducting our research by pivoting from Kim's study and supported by studies on user engagement and format preference, we aim to better understand how public libraries can balance physical reader engagement with digital formats in an increasingly digital landscape.

III. Objectives of the Study

- Discover trends in the number of books, eBooks, and Kindle books published annually.
- Determine the impact of digital borrowing on physical library visits.
- Classify libraries based on their levels of digital participation using machine learning models.
- Suggest strategies for libraries to better adapt to changing consumer behaviors.
- Identify trends in the use of public libraries in Texas from 2018 to 2023.
- Forecast future patterns in print and digital circulation through 2026.
- Classify Texas libraries by level of digital circulation.

- Examine the relationship between Kindle eBook sales and library engagement metrics.
- Assess whether digital publishing is augmenting or replacing traditional library services.

IV. Data Collection and Preprocessing

To begin our analysis, we collected two datasets that represented different but similar aspects of the public library and the digital publishing statistics. The first dataset, obtained from the Texas State Library and Archives Commission (TSLAC), spans from 1996 to 2023. However, for our study we narrowed the scope, and focused specifically on the years between 2018 and 2023 to better reflect recent trends. This dataset includes comprehensive statistics of all Texas public libraries, with key variables such as annual visits, registered users, total circulation, eBook and audiobook checkouts, and material expenditures. The second dataset was sourced from Kaggle and contains Kindle sales data from 2023 and helped in providing insights into digital publishing trends such as sales performance and reader preferences.

In preparation for analysis, we performed several preprocessing steps to refine the raw data and to improve our models' reliability. First, we deleted everything but only the most relevant data (Annual Visits, Registered Users, Total Circulation, Digital Circulation, Print and Electronic Material Expenditures) based on their direct relationship with our research objectives. Next, we renamed the default columns for improved clarity, converted all currency and numeric fields into a more constant format, and removed any repeating or missing values.

Next, we combined the multiple yearly library datasets into a single merged library dataset csv file, ensuring we maintained consistent structure across all our reporting years. We handled missing values through imputation or removal, depending on the frequency and significance of the missing data. Duplicates and constant-value columns were eliminated to prevent bias in our machine learning models. These preprocessing procedures allowed us to create a clean, integrated dataset ready for robust statistical analysis and predictive modeling.

V. Hypotheses for the Study

The research investigates the impact of digital publishing through both Kindle sales and library eBook circulation on conventional metrics related to library engagement like total circulation numbers and visitor traffic. We developed systematic hypotheses which corresponded to the analytical models to study these associations.

- **Primary Hypothesis (H1)**: Statistical evidence shows digital publishing activities measured by Kindle sales together with eBooks and digital circulation volumes create changes in public library engagement metrics including total annual circulation and annual visits.The hypothesis demonstrates that greater digital reading behavior caused by private Kindle buys or expanded library eBook lending corresponds to quantifiable changes in public library community interaction.
- **Null Hypothesis (H0)**: The two variables do not show an aligned association that would affect public library engagement levels. The implementation of digital technology in this situation exists as a standalone procedure that produces insignificant changes to library users' traditional material engagement levels and maintenance trends

Rationale: The proposed assumptions establish that digital transformation affects reader content consumption patterns. Research on Kindle marketplace data and state-level library statistics provides scientists with the ability to study digital behavior changes alongside institutional digital service provision to identify their effect on physical library engagement.

VI. Exploratory Data Analysis (EDA)

A thorough Examining Data Analysis (EDA) investigation of Texas public library (2018–2023) and Kindle marketplace data from 2023 exposed vital patterns needed for our model selection. This phase delivered essential information about usage development together with demographic and digital use variations among different areas.

A. Time Series Trend Analysis

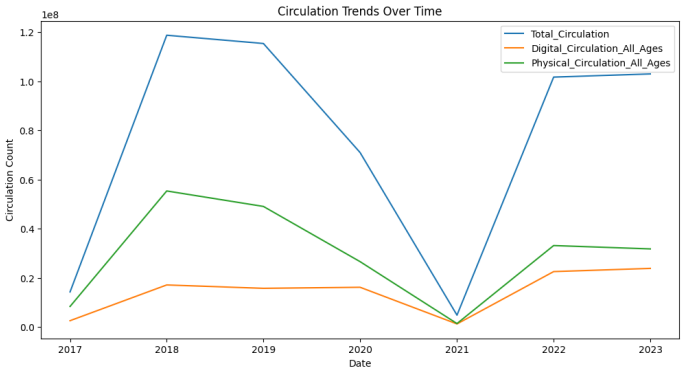

Fig. 1. Circulation trends over time

The analysis started by examining yearly statistical patterns of libraries. The "Circulation Trends Over Time" figure demonstrated how physical book circulation rapidly decreased after 2019 until 2020 and 2021 when COVID-19 created general disruptions. Digital circulation numbers experienced constant growth during this same timeframe which used to demonstrate that increasingly choose to access eBooks and audiobooks digitally.

B. Geographic Circulation Patterns

The Treemap visualization "Total Circulation by Mailing City" illustrated three major contributors namely Houston-HCPL, Austin, and San Antonio. The "Others" grouping in the chart showed how Texas contains different-sized library operations alongside one another.

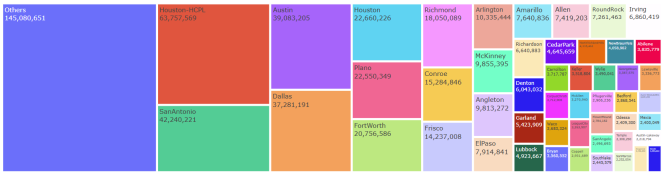

Fig. 2. Circulation distribution by city

C. Population vs. Visits Analysis

Certain libraries showed abnormal patterns in the Legal Service Area Population versus Annual Visits scatter plot because their visit rates deviated strongly from their population figures.

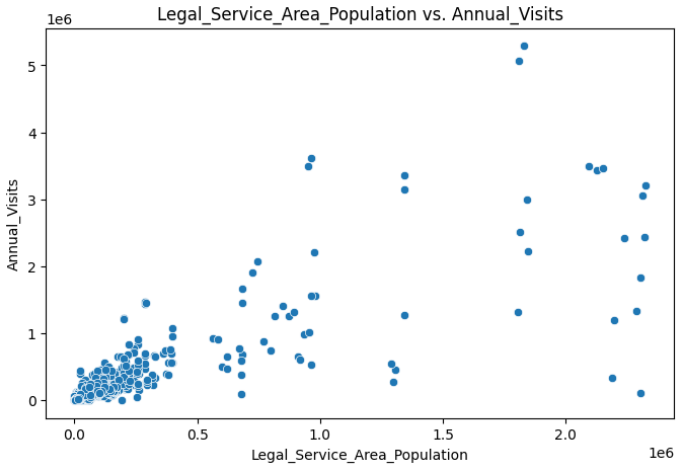

Fig. 3. Population vs. Library Visits

D. Outlier Detection and Treatment

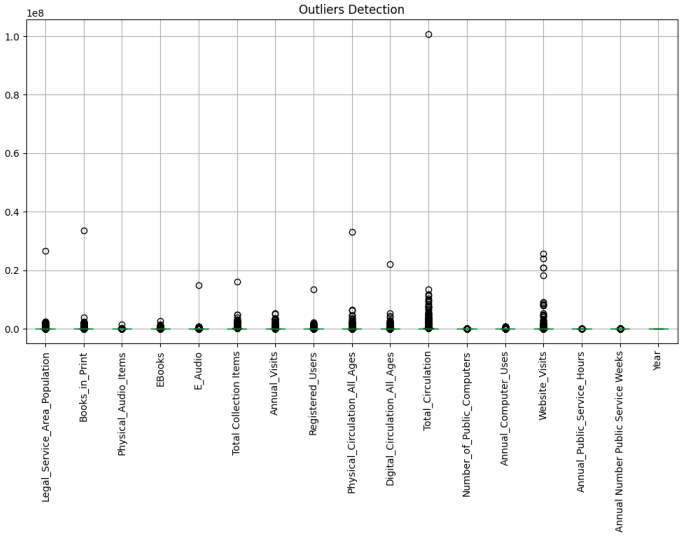

Fig. 4. Outliers before treatment

A robust system was used to identify and handle data quality issues. The boxplots of Total Circulation, Annual Visits and Website Visits numbers indicated large data variability together with identified outliers. The IQR method managed outliers while bettering the entire dataset efficiency through minimizing data skewness. The analysis success rate became evident after implementing post-treatment boxplots.

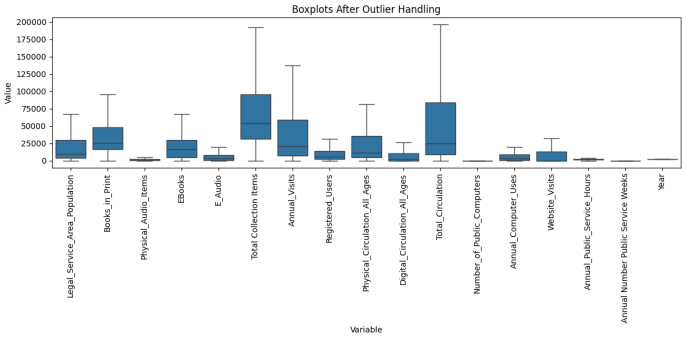

Fig. 5. Outliers after treatment.

E. Correlation Analysis

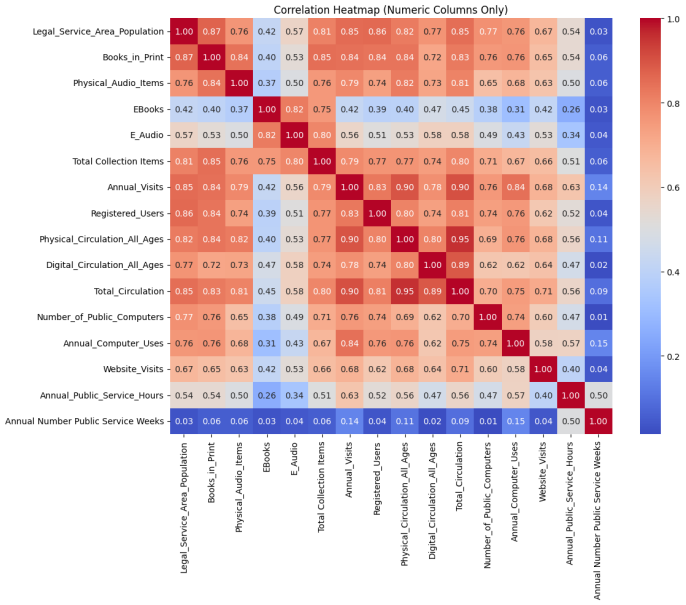

Fig. 6. Correlation Heatmap

The correlation heatmap showed all numeric variables so we could understand their relationships. Total Circulation demonstrated excellent positive correlations with Registered Users (0.89) and Annual Visits (0.90) and all physical and digital circulation variables which indicate their usefulness in predictive modeling processes. The first two principal components from PCA analysis demonstrated their strength as data representatives because they captured more than 93

Visual evaluations of the numerical data distributions were made with histogram plots. In most variables, a pattern that was right-skewed was present that supports the requirement of data normalization before beginning modeling work. Count plot distribution graphs allowed researchers to see how qualitative data such as library types and regions are spread among the data points.

EDA provided a complete data analysis that helped us determine our appropriate models including Prophet and Random Forest and XGBoost and OLS respectively. The models functioned according to the data structure and characteristics.

VII. Data Analytics

A. Model Architecture

Our project framework consists of several phases exist within our project framework for analyzing data, starting from preparation until modeling then visualization. The modeling core required the creation of three analytical models to fulfill our study requirements according to established structural guidelines. Dataset - ("Public Library Accreditation and Statistics", n.d.)

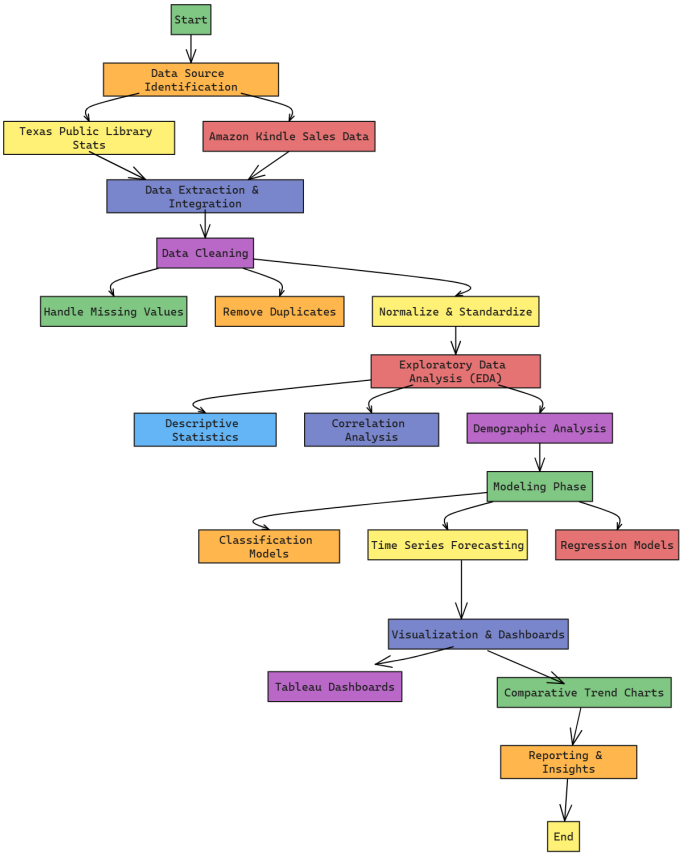

Fig. 7. Project Pipeline

Prophet time series forecasting generated forecasts concerning both physical book circulation and eBook usage and yearly library visits from 2018 to 2026. Prophet received configuration using yearly seasonal patterns combined with trend disruption change points and anomaly detection indicators to handle real-life distribution irregularities during COVID-impacted years 2020 and 2023.

The assessment of digital engagement categories (Low, Medium, High) through model implementation used Random Forest and XGBoost classification techniques for library data and service population characteristics with material usage and digital expenditure analysis. The models discovered general trends within libraries which made it possible to evaluate their digital readiness through comparative analysis.

The study applied Ordinary Least Squares (OLS) regression to evaluate how Kindle sales together with digital content volumes (eBooks and digital circulation) impact total annual circulation and library visit numbers statistically. This analytical model delivered optimum results for both confirming hypotheses and calculating linear correlations between digital elements and customer interactions.

The combination of forecasting and classification models and regression analysis delivered complete, understandable results about digital publishing trends' effect on Texas public library usage patterns.

B. Splitting Data into Train and Test Sets

Our designed architecture required specific training techniques according to the model requirements and analytical requirements and the characteristics of the structural data. Data from 2017 to 2021 served as training data while testing occurred exclusively in 2022-2023 before making forecasts up to 2026 to facilitate strategic decision making. Classification models implemented an 80: 20 stratified approach for train-testing partition to maintain Low, Medium and High digital circulation category proportions. In the regression analysis, we analyzed 2023 data by treating it as distinct time points to study Kindle sales relationships with different library statistics. All training began with scale operations applied to numerical values and then label encoding applied to categorical variables.

```
# Train/Test Split
train_books = data[data['ds'] <= '2021-12-31'][['ds', 'books_y', 'anomaly_books', 'peak']].rename(columns={'books_y': 'y'})
test_books = data[data['ds'] >= '2022-12-31'][['ds', 'books_y', 'anomaly_books', 'peak']].rename(columns={'books_y': 'y'})

train_ebooks = data[data['ds'] <= '2021-12-31'][['ds', 'ebooks_y', 'anomaly_ebooks', 'peak']].rename(columns={'ebooks_y': 'y'})
test_ebooks = data[data['ds'] >= '2022-12-31'][['ds', 'ebooks_y', 'anomaly_ebooks', 'peak']].rename(columns={'ebooks_y': 'y'})

train_visits = data[data['ds'] <= '2021-12-31'][['ds', 'visits_y', 'anomaly_visits', 'peak']].rename(columns={'visits_y': 'y'})
test_visits = data[data['ds'] >= '2022-12-31'][['ds', 'visits_y', 'anomaly_visits', 'peak']].rename(columns={'visits_y': 'y'})
```

```
# Stratified train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

Fig. 8. Train/Test Split

C. Machine Learning Models and Their Application

A combination of time series forecasting functions and service level categorizers and digital publishing effect assessment tools was used in this research.

Prophet (Time Series Forecasting): The forecasting technique was developed by Meta to predict trend changes and seasonal fluctuations and handle holiday effects. We utilized Prophet with three time series changepoints in addition to yearly periods and anomaly detectors set for 2020's COVID-19 decrease and the 2023 post-pandemic alterations. The prediction models operated independently for forecasting Books in Print data alongside eBooks data while generating Library Visits projections. .

Random Forest (Classification): The Random Forest algorithm builds various decision trees when training to yield the most frequent class prediction during classification. With this method we created classifications that identified library preparedness in digital initiatives. The chosen variables for feature selection included the number of digital resources alongside material expenditures and population served. The GridSearchCV method executed the Hyperparameter tuning process.

XGBoost (Classification): XGBoost functions as a high-speed gradient boosting system that uses tree-based learning methods for successive result enhancements in classification tasks. The flexible classification framework combined XGBoost with Random Forest to ensure the stability of the produced results. Numerical datasets benefit from its ability to work with missing information while stopping the model from overfitting which made it an ideal solution for us.

OLS Regression (Linear Regression): The research applied an Ordinary Least Squares regression which used linear regression modeling to study Kindle sales together with digital usage statistics on library total circulation and visits measurements. The analysis employed Kindle Sales figures weighted by region as well as eBook counts and digital item circulations as independent variables. The system generated interpretative results that included statistical coefficients and t-statistics to test hypotheses.

D. Model Evaluation

Each algorithm received its appropriate set of standard evaluation metrics for performance assessment. With Prophet time series forecasting Mean Absolute Error (MAE) together with Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) served for measuring prediction accuracy. The accuracy analysis using confusion matrices validated how well each class was forecasted in the evaluation between Random Forest and XGBoost classification systems. The statistical validity of OLS regression evaluation depended on R-squared for model fit while it required t-tests to determine coefficient significance and an overall F-statistic to validate the model

E. Justification for Model Selection

The selection basis for these models prioritized clear interpretations together with data robustness and applicability to our data type. Prophet served as an ideal time series tool since it performs effectively with recurring patterns and identifies change points in datasets where external events such as pandemic disruptions occur. Random Forest together with XGBoost served as classification models because they excel in processing tabular high-dimensional data and achieve superior results for multi-class predictions. The transparent nature of OLS regression made it an appropriate choice for causal analysis between digital variables and public engagement because it supports hypothesis tests with statistically significant findings

VIII. Results

In an era where digital content dominates and traditional institutions strive for relevance, data tells a compelling story. As libraries transition from quiet reading rooms to digitally equipped knowledge hubs, understanding the metrics behind this transformation becomes essential. This section presents the results of our modeling efforts to quantify and predict how emerging digital platforms—such as Kindle and eBooks—are influencing core library functions including circulation and visits. To do so, we implemented three complementary methodologies: time series forecasting using Prophet, classification through ensemble learning, and regression analysis for quantitative relationship mapping. Each approach contributed uniquely to decoding the interplay between digital publishing and library engagement in Texas from 2018 to 2023.

A. Time Series Analysis

We began by applying the **Prophet** model, a time series forecasting tool, to predict trends in three critical variables: physical book publications, eBook releases, and annual library visits. Prophet's appeal lies in its ability to capture trend, seasonality, and external shocks (e.g., holidays, closures) while maintaining robustness to missing data.

The Time Series model's performance was evaluated through the use of common forecasting metrics, including:

- Mean Absolute Error (MAE)
- Mean Absolute Percentage Error (MAPE)
- Root Mean Squared Error (RMSE)

For physical books, the model yielded:

- MAE: 1.56 million
- RMSE: 1.97 million
- MAPE: 14.35%

These values indicate moderate forecasting accuracy.

In contrast, forecasts for **eBooks** demonstrated higher accuracy:

- MAE: 1.22 million
- RMSE: 1.50 million

These results indicate consistent, linear growth in eBook usage and publication trends.

However, forecasts for **library visits** proved more challenging:

- MAE: 2.93 million
- RMSE: 3.81 million
- MAPE: 25.05%

The volatility in visit behavior likely reflects the impact of pandemic-related disruptions, changing public service hours, and evolving user sentiment regarding in-person visits.

Overall, Prophet effectively captured the steady growth in digital engagement but also underscored the difficulty of modeling human behavioral patterns in a post-pandemic environment.

Metric	Books Forecast	eBooks Forecast	Library Visits Forecast
MAE	1,556,719.59	1,227,885.74	2,934,236.00
RMSE	1,966,087.15	1,497,169.53	3,814,657.93
MAPE (%)	14.35%	14.51%	25.05%
Predicted Values	16.3M, 2.1M	13.5M, 2.0M	15.9M, 1.5M
Actual Values	19.0M, 2.5M	15.6M, 2.37M	21.2M, 1.99M

Fig. 9. Time Series Forecasting Results for Library Metrics

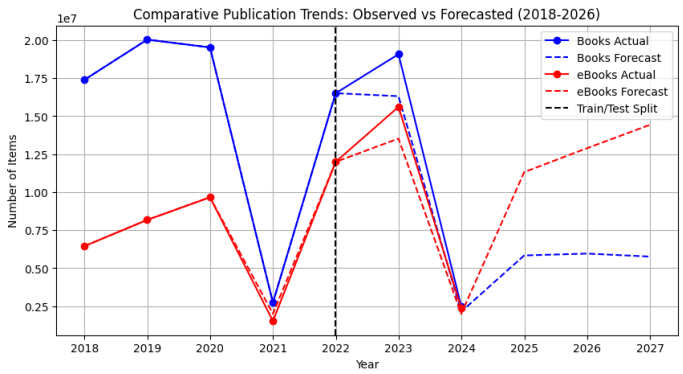

Fig. 10. Time Series Observed vs Forecast graph

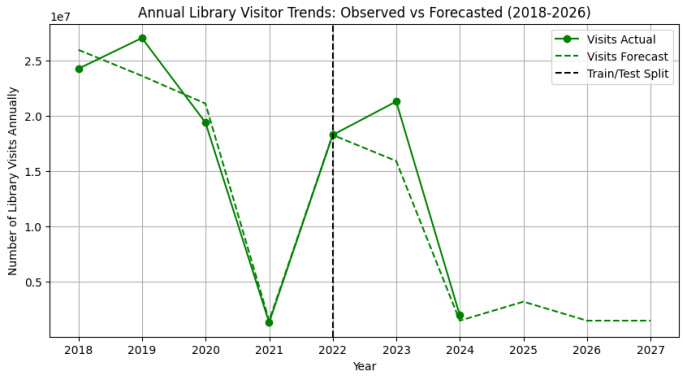

Fig. 11. Time Series Annual Library Visits Trends

B. Classification Model

To segment libraries based on their usage, we developed classification models using supervised machine learning. The goal was to categorize libraries into *Low*, *Medium*, or *High* usage tiers based on total circulation.

We employed **Random Forest** and **XGBoost** classifiers—two ensemble learning methods known for their robustness and accuracy. The training data included features such as:

- Digital Circulation
- Population Served
- eBook Holdings
- Electronic Material Expenditure

To enhance model performance and ensure reliable predictions, we implemented hyperparameter tuning on the Random Forest classifier using GridSearchCV. This approach evaluated different parameter combinations, such as the number of trees (n estimators), tree depth (max depth), minimum samples per leaf (min samples leaf), and minimum samples required to split a node (min samples split). The tuning process involved 3-fold cross-validation to assess each parameter set and select the configuration that produced the highest accuracy.

Afterward, the best-performing combination achieved an accuracy of 86.6 Percent, which was a noticeable improvement over the baseline Random Forest model. This result demonstrates that careful hyperparameter tuning not only improved the classifier's ability to distinguish between low, medium, and high levels of digital circulation but also reduced overfitting by optimizing model complexity. Including this process was crucial to refining our classification model and supports our decision to use Random Forest alongside XGBoost for robust comparative analysis.

To summarize: The Random Forest method achieved an accuracy of **86.5%**, with strong performance in identifying the Medium and Low usage categories. However, XGBoost method was able to marginally outperformed it, reaching **87.6% accuracy** and improving predictions for the High usage class.

Feature importance analysis from the XGBoost model revealed that **Digital Circulation** was the most influential variable, followed by **Population Served**, **eBook Holdings**, and **Electronic Expenditure**.

These insights validate the hypothesis that digital resource availability and community size directly impact library success. The classification framework developed offers a valuable tool for benchmarking performance and guiding strategic development across public library systems.

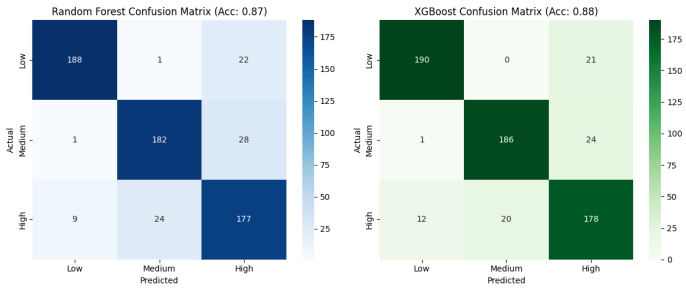

Fig. 12. Confusion Matrices for Random Forest and XGBoost classifiers

C. Regression Analysis

For deeper quantitative insight, we constructed multivariate regression models to measure the relationship between digital publishing trends and core library metrics. Two regression

models were developed: one predicting **Total Annual Circulation**, and another estimating **Annual Library Visits**. we used a new dataset ("Amazon Kindle Books Dataset 2023 (130K Books)", n.d.)

In the first model, the R^2 value reached an impressive **0.937**, indicating that 93.7% of the variance in circulation could be explained by the predictors. **Digital Circulation** was the most impactful variable (coefficient = 2.6986, $p < 0.001$), with **eBooks** also showing significant influence (coefficient = 0.3436, $p < 0.001$). **Kindle Sales**, while statistically significant due to sample size, had a near-zero coefficient (3.311×10^{-16}), confirming it had negligible real-world impact on library circulation.

This points to a crucial finding: *Kindle operates largely in a separate commercial ecosystem and does not directly substitute or enhance library usage*.

```
Model 1: Impact of Kindle Sales on Total Annual Circulation
                            OLS Regression Results
==============================================================================
Dep. Variable:     Total Annual Circulation   R-squared:                       0.937
Model:                                  OLS   Adj. R-squared:                  0.937
Method:                       Least Squares   F-statistic:                     3814.
Date:                      Fri, 02 May 2025   Prob (F-statistic):          2.88e-308
Time:                              16:45:44   Log-Likelihood:                -7076.0
No. Observations:                       515   AIC:                         1.416e+04
Df Residuals:                           512   BIC:                         1.417e+04
Df Model:                                 2
Covariance Type:                  nonrobust
==========================================================================================
                                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                                   5.301e+04   1.07e+04      4.933      0.000    3.19e+04    7.41e+04
Kindle_Sales                            3.311e-16   7.47e-18     44.307      0.000    3.16e-16    3.46e-16
E-books                                    0.3436      0.059      5.785      0.000       0.227       0.460
Annual Circulation: All Other Ages Digital Items    2.6986      0.032     83.691      0.000       2.635       2.762
==============================================================================
Omnibus:                      569.664   Durbin-Watson:                   2.144
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            38480.085
Skew:                           5.063   Prob(JB):                         0.00
Kurtosis:                      44.118   Cond. No.                         inf
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is      0. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

Fig. 13. OLS Model 1: Impact of Kindle sales on Total Annual Circulation

```
Model 2: Impact of Kindle Sales on Annual Library Visits
                            OLS Regression Results
==============================================================================
Dep. Variable:     Annual Library Visits   R-squared:                       0.727
Model:                                 OLS   Adj. R-squared:                  0.726
Method:                      Least Squares   F-statistic:                     682.0
Date:                     Fri, 02 May 2025   Prob (F-statistic):          4.21e-145
Time:                             16:45:44   Log-Likelihood:                -6798.2
No. Observations:                      515   AIC:                         1.360e+04
Df Residuals:                          512   BIC:                         1.362e+04
Df Model:                                2
Covariance Type:                 nonrobust
==========================================================================================
                                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                                   4.737e+04   6267.296      7.558      0.000    3.51e+04    5.97e+04
Kindle_Sales                            8.769e-17   4.36e-18     20.122      0.000    7.91e-17    9.62e-17
E-books                                    0.0495      0.035      1.428      0.154      -0.019       0.117
Annual Circulation: All Other Ages Digital Items    0.6707      0.019     35.669      0.000       0.634       0.708
==============================================================================
Omnibus:                      345.236   Durbin-Watson:                   2.249
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            31683.875
Skew:                           2.062   Prob(JB):                         0.00
Kurtosis:                      41.204   Cond. No.                         inf
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is      0. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

Fig. 14. OLS Model 2: Impact of Kindle sales on Total Annual Visits

The second regression model—focused on predicting **Annual Library Visits**—had a slightly lower R^2 score of **0.727**, but still offered meaningful insights. Digital Circulation again emerged as a strong predictor (coefficient = 0.6707, $p < 0.001$), underscoring its importance in drawing readers to physical libraries, possibly for hybrid services like curbside pickup or digital device access. eBook availability, however, was not statistically significant ($p = 0.154$), and Kindle Sales again demonstrated no substantive effect.

This distinction suggests that while having eBooks in a catalog is important, *actual usage*—especially digital lending activity—is what drives library engagement.

IX. Tableau Dashboard and Visual Analytics

A comprehensive Tableau dashboard serves two purposes: it provides stakeholders with interactive library trends analysis as well as enhances interpretation of our analysis. The dashboard merges Texas public library data from multiple years to display essential metrics including circulation statistics and collection information combined with visit data and geographical locations. This research provides parallel insights at the macro and micro level about library transformation since the digital shift occurred between 2018 and 2023.

Bubble Chart: Library Performance

The visual display presents an aggregated view of city circulation levels. Circulation volume distribution appears through bubble size, together with circulation size levels noted by color intensity. Houston, alongside Dallas and Austin, acts as a major population centers, which take the lead position in this chart because of their efficient library system. Both digital and physical engagement, along with resource allocation effectiveness, can be identified by examining this chart at its geographic point of view.

Line Graph: Annual Library Visits Over Time

The time-series graph demonstrates the transformation of library visits between 2018 at early stages and late 2023. The sharp decline in 2020 to 2021 aligns with COVID-19 public closures and the pandemic. The post-2021 recovery period demonstrates a step-by-step improvement in library traffic without consistent patterns which demonstrates users have adopted new patterns of space engagement. Our Prophet model confirms with visual evidence that external shocks create unpredictable visit patterns for the library.

Pie Chart: Physical & Digital Collection Comparison

The illustration examines the yearly modifications in library material distribution patterns during the specified period. Each year displays two corresponding values: the counts for physical and digital collections. Digital content materials exceeded physical items by surpassing 6.77 million collections to 3.49 million in the year 2023. The data from 2018 and 2019 demonstrates the opposite trend because digital holdings grew very quickly.

Geographic Map: Library Metrics

The spatial visual element of the map displays hundreds of Texas libraries throughout the region. A distinct collection performance and service distribution system can be seen through each colored dot or ring that represents individual libraries. This view enables the identification of regional patterns while revealing the digital disparities which mainly appear between urban and rural areas. The classification model high-usage areas in Texas correspond to communities with

denser library clusterings and more developed connectivity patterns in Central and Eastern Texas regions.

Line Chart: Shifting Priorities in Collection Type

This "Shifting Priorities" line chart positions physical book collection and digital material collection rankings starting from 2018 up to 2023. Physical collection rankings decreased throughout the same period as digital collection priorities experienced extensive growth since 2020. Libraries value digital resources more than physical ones as their essential holdings during the year 2023. Digital circulation rates matter most to user engagement because we observe this dual-axis ranking approach in practice.

In conclusion, By using Tableau the numerical data becomes a clear visual presentation for easy interpretation. The solution enables exploration combined with strategic planning functions which connects between raw numbers and actionable information. Users understand Texas library evolution along with digital changes in public access to knowledge by applying interactive elements to visible filters and user-friendly design.

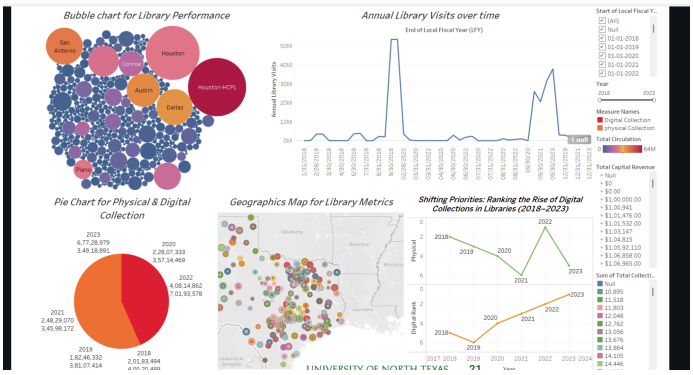

Fig. 15. Tableau Dashboard

Link to Interactive Tableau Dashboard

X. Limitations

A. Limited Geographical Scope

We limited our study to Texas public libraries. The study results lack broad applicability because digital usage patterns and adoption rates differ among states and regional areas outside of Texas.

B. PCA (Principal Component Analysis)

We utilized PCA as an analysis method in our work because it simplifies feature dimensions and controls issues related to feature associations. The application of PCA enhanced efficiency while targeting relevant variance yet presented several critical disadvantages to the method. The application of PCA generates hard-to-understand main components from features and fails to reveal non-linear patterns while neglecting class information which might reduce results quality and remains highly affected by feature normalization and data completeness.

C. Classification Model Constraints

The Random Forest and XGBoost models require high-quality features and balanced outcomes for their application. The classification models show reduced effectiveness in generalizing across various library systems when operating with minimal outcome groups and sparse labeled data.

D. Data Gaps and Missing Values

Annual datasets contained entries that were either fully missing or displayed inconsistent values. The imputation and cleaning mechanisms we applied resolve missing data issues yet their presence affects model accuracy and generalization capabilities.

XI. Future Scope

Future research must expand its analysis to encompass public libraries throughout different states or regions to develop findings that represent a wider range of populations. The analysis of population characteristics including age brackets along with income and occupational profiles provides better understanding about digital and physical reading patterns between different demographic segments. Researchers should examine the power commercial digital platforms Kindle, Audible and Google Books hold in order to determine their effects on public library patronage. Future research should focus on understanding how the evolving digital tools alongside AI-powered search engines affect both reader behavior and library traffic. The research needs to analyze the perfect combination between digital and physical library services to guide strategic investments that develop hybrid service models which address current and evolving user needs.

XII. Conclusion

Overall, our findings indicate a clear behavioral shift. Readers are increasingly choosing digital platforms, which has prompted libraries to expand their eBook and audiobook offerings. Although physical visits have slightly declined, digital circulation has seen a gradual increase, in large part due to digital checkouts. Moreover, from both a policy and management perspective, the results suggest that public libraries can thrive alongside commercial digital platforms like Kindle. Rather than acting purely as competitors, these platforms signal consumer demand and open pathways for libraries to adapt and innovate. The solution appears to be maintaining a hybrid model with both digital and physical books, though determining the best ratio between the two formats would need more research. Strategic investments in digital infrastructure and user experience will be key to maintaining relevance and reach in the coming decade.

Furthermore, this study provides a starting off point for future research and decision-making as the digital shift continues. Continued monitoring of digital engagement metrics, paired with public feedback and usage analytics, will help ensure that libraries remain both resourceful and popular in a changing information landscape. Additionally, future iterations of this paper could look into expanding the library datasets

to include additional states or different regional systems, investigating how different types of online publishing platforms influence library lending trends, or even incorporating reader demographic variables, such as age and occupation to determine how these factors influence format preferences across both digital and physical media. Alternatively, new studies could also explore how research through search engines like Google, and various AI tools have changed reading habits and library visits over time.

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